

Emergency Medicine
For Practical Examination Purpose

Table of contents:

1. AMI
2. Status asthmaticus
3. Diabetic ketoacidosis (DKA)
4. Hyperosmolar non-ketotic coma (HONC)
5. Organophosphorus poisoning
6. Snake bite
7. Thyroid storm
8. Hemoptysis.

AMI

Suspected MI

- a. Airway, breathing and circulation secured
- b. Immediate interventions in case of complications
- c. Loading dose of aspirin 300 mg + Clopedogrel 300 mg
- d. Antiplatelet drug
- e. Atorvastatin (Plaque stabilizer)
- f. Antianginal (Nitrate IV infusion)
- g. Analgesic (Morphine/ Pethidine).

Confirmed MI

STEMI (Full thickness MI)

NSTEMI/ ACS with no MI (Subendocardial MI)

Primary PCI
indicated?

Conservative
treatment*

Yes

No

PCI + Stenting

Thrombolysis
indicated?

Yes

No

Thrombolysis

Conservative
treatment*

If successful

If fails

Conservative
treatment*

Rescue PCI with stenting

Conservative
treatment*

***Conservative treatment**

It consists of 2 parts:

- a. Risk factor modification.
- b. Specific treatment.

Risk factor modification

- A.** ABDOMINAL OBESITY: Regular exercise and reduction of abdominal fat.
- B.** BLOOD LIPID ABNORMALITY: Regular monitoring and control of LDL levels.
- C.** CIGARETTE: Reduction/ cessation of cigarette smoking.
- D.** DIABETES: Strict monitoring and control of blood sugar level.
- E.** EXCESS ALCOHOL: Avoidance/ complete cessation of alcohol ingestion.
- H.** HYPERTENSION: Treat hypertension adequately and monitor BP on a regular basis.

Specific treatment

1. Antiplatelet:
 - Aspirin: Lifelong.
 - Clopidogrel/ Prasugrel/ Ticagrelor: Continued for 9-12 months after stenting (upto 24 months).
2. Anticoagulant:
 - Unfractionated heparin.
 - Low molecular weight heparin (LMW-Heparin).
 - Fondaparinux.
3. Atorvastatin/ any other statins.
4. ACE inhibitor/ ARB.
5. Anti-anginal agents: Nitrate/ Nicorandil/ Ranolazine.
6. Beta blocker: Metoprolol/ Bisoprolol/ Carvedilol.

STATUS ASTHMATICUS

A. Airway:

Protected by frequent suction, intubation if required.

B. Breathing:

- ✓ High flow oxygen (target saturation > 95%)
- ✓ Assisted ventilation: Non-invasive (CPAP)/ Invasive.

C. Circulatory support by IV fluid

D. Drugs:

A. Antibiotic: Short course

A. Aminophylline infusion:

If patient remains symptomatic after optimal therapy.

B. Bronchodilator:

Nebulization with (Salbutamol + Budesonide + Ipratropium [**SBI**]):

Initially repeated every 15-30 minutes. When acute stage is over, it is given every 4-6 hours till the patient is stable enough to use inhaler.

C. Corticosteroids:

Oral (Prednisolone)/ IV (Hydrocortisone) for 5-7 days.

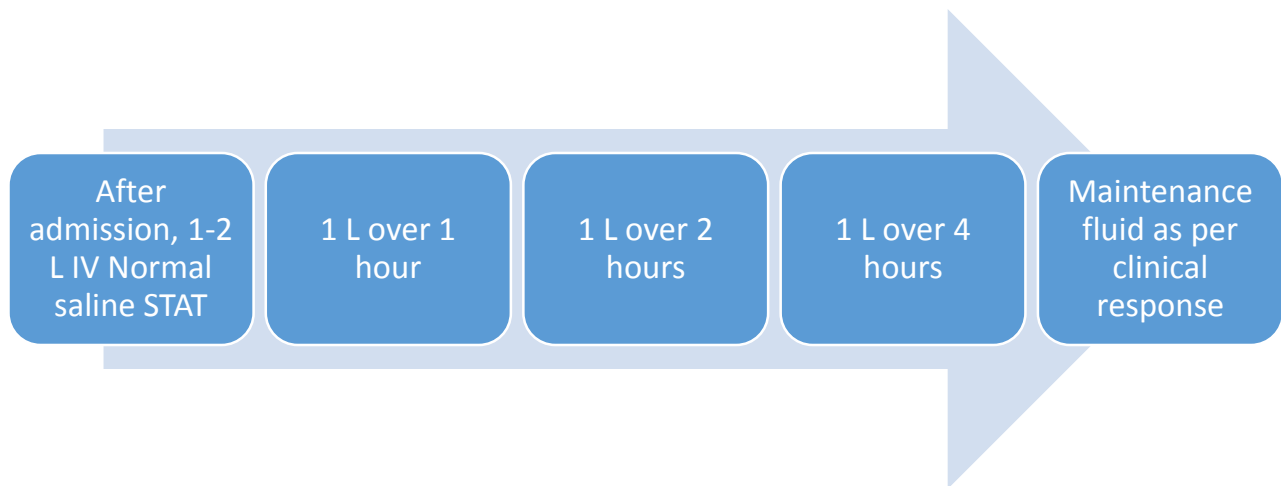
M. A single dose of IV MgSO₄.

DKA

I. Dehydration:

Aggressive IV fluid therapy.

In DKA, there is often 6-8 L fluid deficit; however, amount of fluid to be replaced depend on the patient's clinical status (i.e. severity of dehydration).



II. K⁺ supplementation:

To be replaced (as IV insulin will rapidly shift K⁺ to intracellular space causing serum hypokalemia): Usually KCl is added to IV fluid bottle.

III. Anti-hyperglycemic agents:

- IV insulin infusion, usually regular insulin is given through an infusion/ syringe pump: the rate of infusion is titrated up/ down according to hourly CBG result.
- Insulin infusion is usually continued till the patient is critically ill and usually switched over to SC insulin once the patient is fit enough to eat and drink properly.
- Once CBG remains consistently <200 mg/dl, normal saline should be replaced by 5% dextrose normal saline.

IV. Acidosis:

- Does not usually require any specific intervention as correction of hyperglycemia will gradually lead to disappearance of ketoacidosis and ketonuria.
- Specific intervention like IV NaHCO₃ rarely required, only when serum pH is dangerously acidotic (<6.9).

V. Prophylactic anticoagulant:

Unfractionated heparin, LMW heparin: to prevent DVT; as severe dehydration is an important risk factor causing hemo-concentration.

VI. Broad spectrum antibiotic:

If any clinical suspicion of infection.

VII. Subsequent treatment once blood sugar is stable and patient starts to eat and drink normally:

- ✓ Stop IV fluid
- ✓ Switch over to SC insulin
- ✓ Outline a long-term management plan for DKA.

HONC

I. Dehydration:

Aggressive IV fluid therapy: fluid deficit usually 8-10 L



- II. K⁺ supplementation
- III. Anti-hyperglycemic agents
- IV. Anticoagulant
- V. Antibiotic

Same as DKA

VI. Subsequent treatment: Oral hypoglycemic agent $\pm$ Insulin.

ORGANO-PHOSPHORUS POISONING

- A.** Airway: To be protected, if required intubate particularly if profuse airway secretions
- A.** Admit to ICU/ ITU
- B.** Breathing support: O₂/ Ventilatory support, often required
- C.** Circulation: IV fluid

C. Catheterize- Monitor urine output.

D. Decontamination:

- ✓ Patient to be washed with copious amount of water and soap
- ✓ Ocular decontamination- Washing eyes with water/ normal saline
- ✓ Gastric Decontamination- Gastric lavage.

D. Drugs-

1. Atropine: Antidote (Mainstay of Rx):

IV Atropine should be started immediately: Loading bolus dose 1-2 mg depending on severity and then to repeat same boluses every 5 minutes (in doubling doses in case no improvement is seen) till atropinisation is achieved.

Target end points for Atropine therapy:

- ✓ Clear chest on auscultation with no wheeze
- ✓ Heart rate > 80 beats/min
- ✓ Pupil no longer pin point
- ✓ Dry axilla
- ✓ Systolic blood pressure > 80 mm of Hg.

Once atropine maintenance target end point is achieved, often required to be maintained by atropine infusion.

2. Pralidoxime- as adjunct to atropine

3. Diazepam- to sedate/to control convulsion.

E. Evidences to be preserved for Police- Gastric lavage sample/urine sample/any suicide note found

F. File police report (Very important... never forget).

SNAKE BITE

1. First aid:

- A. Immobilize the victim's bitten limb using a splint and lightly put a bandage.
- B. Be prepared to treat the shock and provide cardiopulmonary resuscitation (CPR)
- C. **Do not** apply a tourniquet/ wash the bite site with soap or any other solution to remove the venom/ make cuts or incisions on or near the bitten area/ attempt to suck out venom with your mouth.

2. Definitive Rx:

A- Always, if possible the snake is to be identified. Victim's relatives can be asked to bring the snake to **hospital (if it has been killed)** to identify it. Exact time of bite to be determined.

A- Airway- To be secured, if required intubate

A- Admit ALL and observe for atleast 24 hours (onset of effects of envenomation may be delayed)

B- Breathing- O₂/ if required Ventilate particularly those with respiratory failure

C- Circulation- Treatment of shock- Initial IV fluid resuscitation/
Vasopressors/Maintenance IV fluid

C- 20 minute whole blood Clotting test (20WBCT) ** - A few mls of fresh venous blood should be placed in a clean and dry **glass vessel** preferably test tube and left undisturbed at ambient temperature for 20 minutes. After that tube should be gently tilted to detect whether blood is still liquid and if so then blood is incoagulable. The test should be carried out every 30 minutes from admission for 3 hours and then hourly after that (****important clinically**).

D-Dialysis- for AKI

Drugs-

1. Tetanus toxoid if skin is breached
2. Antibiotics if there is cellulitis/ local necrosis
3. Analgesics- Paracetamol +/- Tramadol
4. Neostigmine- Effective in postsynaptic neurotoxins such as those of Cobra

and is not useful against presynaptic neurotoxin i.e. common Krait and the Russell's viper

5. **Anti-snake venom (ASV):**

Anti-snake Venom Dose

Initial Dose:

- ✓ Mild envenomation= Neurotoxic/Hemotoxic Systemic symptoms manifest > 3 hours after bite- **8–10** Vials
- ✓ Severe envenomation= Neurotoxic/Hemotoxic Systemic symptoms manifest < 3 hours after bite- **8** Vials

(Children- same ASV dosage as adults)

Repeat Doses:

After initial ASV dose, no additional ASV should be given until the next clotting test at **6 hours**. This is due to the inability of the liver to replace clotting factors in less than 6 hours.

If WBCT is >20 minutes; repeat dose of **5–10** vials of ASV can be given and should be continued 6 hourly till coagulation is restored/ species is identified against which polyvalent ASV is ineffective.

THYROID STORM

1. Large doses of propylthiouracil (600 mg loading dose and 200-300 mg every 6 h): Orally/ by nasogastric tube/ per rectum
2. 1 hour after the first dose of propylthiouracil, stable iodide is given to block thyroid hormone synthesis
3. A saturated solution of potassium iodide: Orally
4. Propranolol: To reduce tachycardia and other adrenergic manifestations (40–60 mg PO every 4 h/ 2 mg IV every 4 h)
5. Glucocorticoids (Dexamethasone)
6. Antibiotics (if infection is present)
7. Cooling, oxygen and IV fluids.

HEMOPTYSIS

1. Complete bed rest
2. Chest X Ray
3. Injection Diazepam
4. IV drip of normal saline
5. Record vital signs repeatedly
6. Cough suppressant: Linctus Codeine
7. Injection Ethamsylate
8. Syrup Amoxicillin
9. Stop bleeding by tamponade effect: By applying a Fogarty balloon catheter
10. Bronchial/ Pulmonary artery embolization: Last resort in uncontrollable cases
11. If respiratory distress starts, send the patient to Respiratory Care Unit (RCU).